The complete enumeration of strictly Deza graphs with at most 21 vertices

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Based on joint project with S. Goryainov and L. Shalaginov

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Strongly regular graphs

A k-regular graph on v vertices is called a strongly regular graph with parameters (v, k, λ, μ), if any two adjacent vertices in this graph have λ common neighbours and any two distinct non-adjacent vertices have μ common neighbours.

Deza graphs

Deza graphs were introduced in 1999 [1] as a generalisation of strongly regular graphs.

A k-regular graph on v vertices is called a Deza graph with parameters (v, k, b, a), $b \geq a$ if the number of common neighbours of any two distinct vertices takes two values: a or b.

A Deza graph is called a strictly Deza graph if it has diameter 2 and is not strongly regular.

[1] M. Erickson, S. Fernando, W.H. Haemers, D. Hardy, J. Hemmeter, Deza graphs: A generalization of strongly regular graphs, *J. Comb. Des.*, **7** no. 6, (1999) 359–405

Parameters of Deza graphs

Let N_v be the neighbourhood of vertex v: $N_v = \{u : u \sim v\}$.
For fixed vertex u of Deza graph Γ we define
$\alpha = |\{v \in \Gamma : |N_u \cap N_v| = a\}|$,
$\beta = |\{v \in \Gamma : |N_u \cap N_v| = b\}|$.

Lemma 1 ([1, Proposition 1.1])

α and β do not depend on u and

$$\alpha = \begin{cases} \dfrac{b(n-1) - k(k-1)}{b-a}, \textit{if } a \neq b, \\ \dfrac{k(k-1)}{a}, \textit{if } a = b; \end{cases}$$

$$\beta = \begin{cases} \dfrac{a(n-1) - k(k-1)}{a-b}, \textit{if } a \neq b, \\ \dfrac{k(k-1)}{a}, \textit{if } a = b. \end{cases}$$

Parameters of Deza graphs

Lemma 2 ([1, Corollary 1.2])

Let Γ be Deza graph with parameters $(v, k, b, a), \beta, \alpha$. Then
(1) $b - a$ divides $b(n-1) - k(k-1)$;
(2) if $\alpha \neq 0$, then $n \geq 2k - a$;
(3) if $\alpha, \beta \neq 0$, then $a(n-1) < k(k-1) < b(n-1)$.

In 1999 the complete list of strictly Deza graphs with at most 13 vertices was presented (see [1]).
In 2011 an algorithm for enumeration of Deza graphs was presented by S. Goryainov and L. Shalaginov and this list was extended up to 16 vertices (see [2]).

[2] S. V. Goryainov, L. V. Shalaginov, On Deza graphs with 14, 15, and 16 vertices, *Sib. Electron. Mat. Izv.*, **8**, (2011) 105–115

Original algorithm: preparations

The first step is to obtain the list of possible parameters of Deza graphs with the restrictions from Lemma 2.

Then the algorithm produces adjacency matrices of Deza graphs with required parameters, where each row represented as single integer number in binary form.

To accomplish that we construct three special arrays before we start the enumeration of matrices:

$NumberOfOnes$, one-dimensional array which contains the number of 1s in binary form of number n for each n;

$NumberOfCombinations$, two-dimensional array which contains the number of combinations $\binom{n}{k}$ for each n and k;

$Combinations$, three-dimensional array which contains all possible combinations of k 1s in n-digit binary number for each n and k.

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10------  <---- Combinations[6][3] = 111000     011100
1-0-----                             110100     011010
1--0----                             110010     011001
1---0---                             110001     010110
0----0--                             101100     010101
0-----0-                             101010     010011
0------0                             101001     001110
                                     100110     001101
                                     100101     001011
                                     100011     000111
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10111000  <---- Combinations[6][3] = 111000 !   011100
110-----                             110100     011010
11-0----                             110010     011001
11--0---                             110001     010110
00---0--                             101100     010101
00----0-                             101010     010011
00-----0                             101001     001110
                                     100110     001101
                                     100101     001011
                                     100011     000111
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100  <---- Combinations[6][3] = 111000 X   011100
110-----                             110100 !   011010
11-0----                             110010     011001
10--0---                             110001     010110
01---0--                             101100     010101
00----0-                             101010     010011
00-----0                             101001     001110
                                     100110     001101
                                     100101     001011
                                     100011     000111
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100
110-----  <---- Combinations[5][2] = 11000     01010
11-0----                             10100     01001
10--0---                             10010     00110
01---0--                             10001     00101
00----0-                             01100     00011
00-----0
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100
11010010  <---- Combinations[5][2] = 11000 X   01010
1110----                             10100 X   01001
100-0---                             10010 !   00110
010--0--                             10001     00101
001---0-                             01100     00011
000----0
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100
11010010
1110----  <---- Combinations[4][1] = 1000    0010
100-0---                             0100    0001
010--0--
001---0-
000----0
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100
11010010
11100001  <---- Combinations[4][1] = 1000 X   0010 X
10000---                            0100 X   0001 !
0100-0--
0010--0-
0001---0
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100
11010010
11100001
10000---  <---- Combinations[3][3] = 111
0100-0--
0010--0-
0001---0
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100
11010010
11100001
10000---  <---- Combinations[3][3] = 111 X
0100-0--
0010--0-
0001---0
```

Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100
11010010
1110----  <---- Combinations[4][1] = 1000 X   0010 X
100-0---                             0100 X   0001 X
010--0--
001---0-
000----0
```

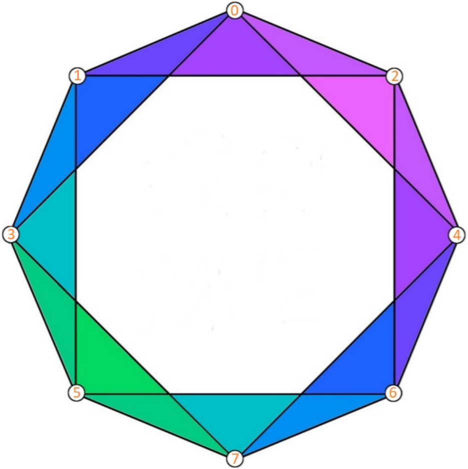
Original algorithm: example

```
parameters: (8, 4, 2, 1) beta = 5, alpha = 2

01111000
10110100
11010001  <---- Combinations[5][2] = 11000 X   01010
1110----                             10100 X   01001
100-0---                             10010 X   00110
010--0--                             10001 !   00101
000---0-                             01100     00011
001----0
```

Original algorithm: example

01111000
10110100
11001010
11000101
10100011
01010011
00101101
00011110



Improvements: the first two rows

Let us consider three possible cases:

1. $\alpha < k$. Then we can find two adjacent vertices which have exactly b common neighbors. In that case the first two rows of the adjacency matrix look like this:

$$
\begin{array}{l}
01\ \overbrace{1\ldots1}^{b}\ \overbrace{1\ldots1}^{k-b-1}\ \overbrace{0\ldots0}^{k-b-1}\ \overbrace{0\ldots0}^{n-2k+b} \\
10\ 1\ldots1\ 0\ldots0\ 1\ldots1\ 0\ldots0
\end{array}
$$

2. $\alpha = k$. Then we can find two adjacent vertices which have exactly b common neighbors, because we obtain a strongly regular graph when we can't find such vertices. In that case the first two rows of the adjacency matrix look the same as in case 1.

3. $\alpha > k$. Then we can find two non-adjacent vertices which have exactly a common neighbors. In that case the first two rows of the adjacency matrix look like this:

$$
\begin{array}{lllll}
00 & \overbrace{1\ldots1}^{a} & \overbrace{1\ldots1}^{k-a} & \overbrace{0\ldots0}^{k-a} & \overbrace{0\ldots0}^{n-2k+a-2} \\
00 & 1\ldots1 & 0\ldots0 & 1\ldots1 & 0\ldots0
\end{array}
$$

Improvements: the next rows

Assume we have the first two rows of the adjacency matrix and trying to construct the third row:

```
011  11111100000
101  11100011100
110  -----------
```

Notice that columns can be divided into four blocks

```
011  111|111|000|00
101  111|000|111|00
110  ---|---|---|--
```

and changing the order of 1s inside each block gives an equivalent matrix.

Improvements: the next rows

For example these two matrices are equivalent:

```
011   111|111|000|00        011   111|111|000|00
101   111|000|111|00        101   111|000|111|00
110   110|100|110|10        110   101|001|011|10
```

For the next rows, this division into blocks can be extended, but with each row the number of blocks multiplies by 2 (8 blocks for the 4th row, 16 blocks for the 5th row, etc.).

Remarks

- We use Magma to check whether graphs are isomorphic after the completion of the enumeration. We also use Magma after adding the 3rd, the 4th and the 5th row.
- In case of $a = 0$ we use Magma to calculate the diameter of the resulting graphs. If the diameter does not equal 2, then this graph is not strictly Deza graph.

Table of parameters

Parameters	Total	VT Graphs	Cayley Graphs	From Association Schemes
(8, 4, 2, 0)	1	1	1	0
(8, 4, 2, 1)	1	1	1	1
(8, 5, 4, 2)	1	1	1	1
(9, 4, 2, 1)	2	1	1	1
(10, 5, 4, 2)	1	1	1	1
(12, 5, 2, 1)	1	1	1	1
(12, 6, 3, 2)	2	1	1	1
(12, 7, 4, 3)	1	1	1	1
(12, 7, 6, 2)	1	1	1	1
(12, 9, 8, 6)	1	1	1	1
(13, 8, 5, 4)	1	1	1	1
(14, 9, 6, 4)	1	1	1	1
(15, 6, 3, 1)	1	0	0	0
(16, 5, 2, 1)	1	1	1	0
(16, 7, 4, 2)	2	2	2	2
(16, 8, 4, 2)	1	1	1	1
(16, 9, 6, 4)	2	1	1	1
(16, 9, 8, 2)	1	1	1	1
(16, 11, 8, 6)	1	1	1	1
(16, 12, 10, 8)	1	1	1	1
(16, 13, 12, 10)	1	1	1	1
(17, 8, 4, 3)	3	0	0	0

Parameters	Total	VT Graphs	Cayley Graphs	From Association Schemes
(18, 8, 4, 2)	7	2	2	1
(18, 8, 4, 3)	1	0	0	0
(18, 9, 6, 4)	2	1	1	1
(18, 9, 8, 4)	2	1	1	1
(18, 13, 12, 8)	1	1	1	1
(19, 6, 2, 1)	6	1	1	1
(19, 8, 4, 2)	3	0	0	0
(19, 12, 8, 7)	2	0	0	0
(20, 6, 2, 1)	37	1	0	1
(20, 7, 3, 2)	1	1	1	1
(20, 7, 6, 2)	2	1	1	1
(20, 8, 4, 2)	5	1	1	0
(20, 10, 6, 4)	6	1	1	1
(20, 11, 10, 2)	1	1	1	1
(20, 13, 9, 8)	1	1	1	1
(20, 13, 12, 8)	1	1	1	1
(20, 14, 10, 9)	1	1	0	1
(20, 17, 16, 14)	1	1	1	1
(21, 8, 3, 2)	3	1	1	1
(21, 8, 4, 2)	19	1	1	1
(21, 10, 5, 4)	4	0	0	0
(21, 10, 6, 3)	1	0	0	0
(21, 12, 7, 5)	2	1	1	1
(21, 12, 7, 6)	2	2	2	2

Cayley graphs

Let G be a group and S be a set with the property: $S = S^{-1}$, that is $\forall s \in S \; \exists s^{-1} \in S$ and $e \notin S$.
Cayley graph $Cay(G, S)$ of the group G with the generating set S is a graph whose vertices are elements of the group G and the element g is connected by an edge exactly to those elements that are obtained by multiplying g by elements from S.

Symmetric association schemes

Let X be a set of size n, and $R_0, R_1, \ldots, R_d$ be relations defined on X. Let $A_0, A_1, \ldots, A_d$ be the 0-1 matrices corresponding to the relations, that is, the (x, y)-entry of A_i is 1 if and only if $(x, y) \in R_i$. Then $(X, \{R_i\}_{i=0}^{d})$ is called a d-class symmetric association scheme if

(1) $A_0 = I$;
(2) $\Sigma_{A_i} = J$;
(3) each A_i is symmetric;
(4) for each pair i and j, $A_i A_j = \Sigma_k p_{ij}^k A_k$ for some constants p_{ij}^k.

Deza graphs from symmetric association schemes

Let $(X, \{R_0, R_1, \ldots, R_d\})$ be a symmetric association scheme, and $F \subset \{1, 2, \ldots, d\}$. Let G be the graph with adjacency matrix $\Sigma_{f \in F} A_f$. Then G can be a Deza graph.

Thanks for your attention!